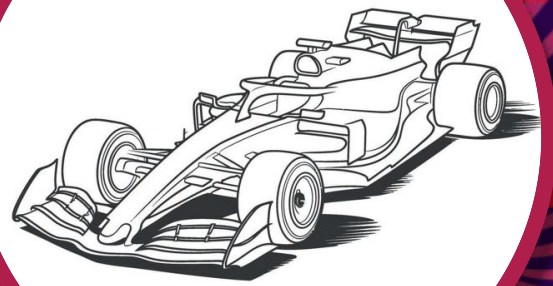
PORTFOLIO

Nattan NEPPER



ABOUT ME



Currently in 5th year at ESTACA and at LINKÖPING UNIVERSITY, in automotive section, I've always been interested about motorsport and the engineering that accompanies it. Throughout my career and experiences, I have sought to combine rigor, technical expertise, curiosity and a spirit of achievement. This portfolio showcases the projects, skills and internships that reflect my commitment to the motorsport, with a particular focus on aerodynamics, mechanical design and vehicle dynamics.

SOMMAIRE

- **Skills**
- **School and SELF projects**
  - -Create a mini wind tunnel
  - -Redesign a range of road trailers
  - -Sizing of a rear suspension for a Formula vee
  - -CFD Simulation of a DRIVAer Car model
  - -CFD simulation of a fire in an open area
  - -Creating the control for an inverted pendulum
  - -Sizing a battery pack
  - -CFD TEST OF A PORSCHE 911 992 GT3 RS WING
  - -Analysing the movement of an engine
- **Intership:**
  - -Metrology internship
  - -Automotive mechanics internship
  - -Design internship for automotive testing
- **And now ?**
- **My Interests**

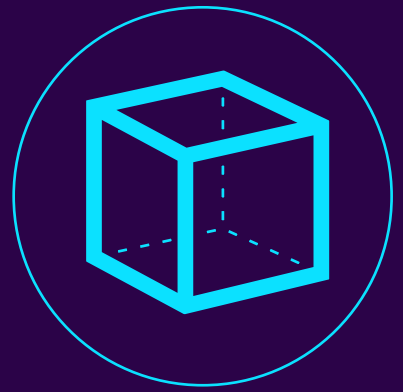
SKILLS



AERODYNAMICS CFD

Flow analysis to optimize performance and support

ANSYS FLUENT
ANSA
PARAVIEW



CAD

3D modeling of parts and assembly under technical constraints

SOLIDWORKS
3DEXPERIENCE
ONSHAPE



VEHICLE MECHANICS

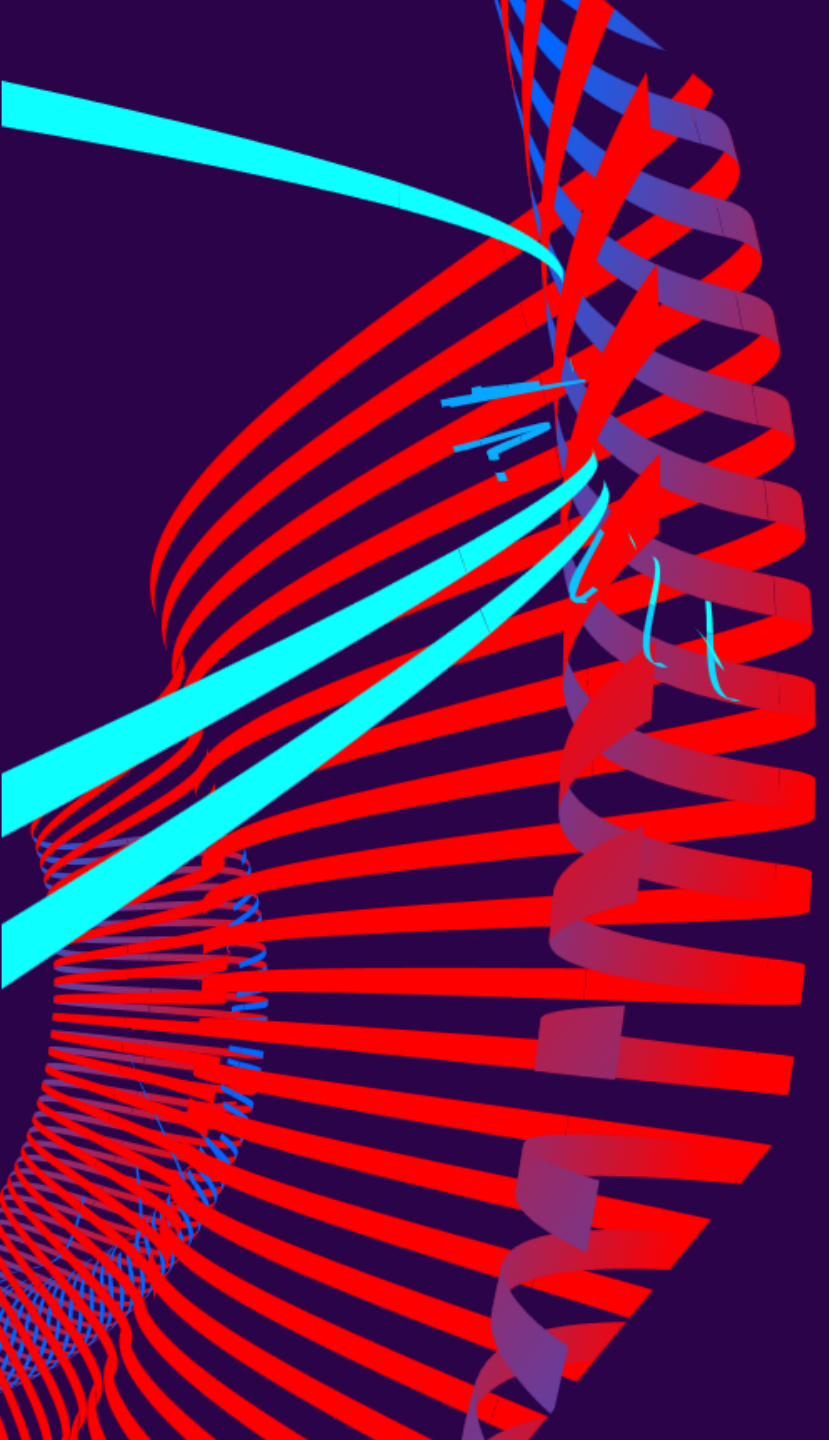
Study of dynamic behavior and design of rolling systems

ADAMS CAR
SIMULINK



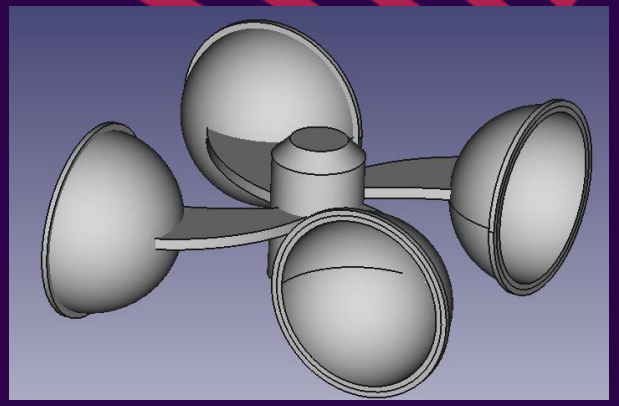
MOTORSPORT

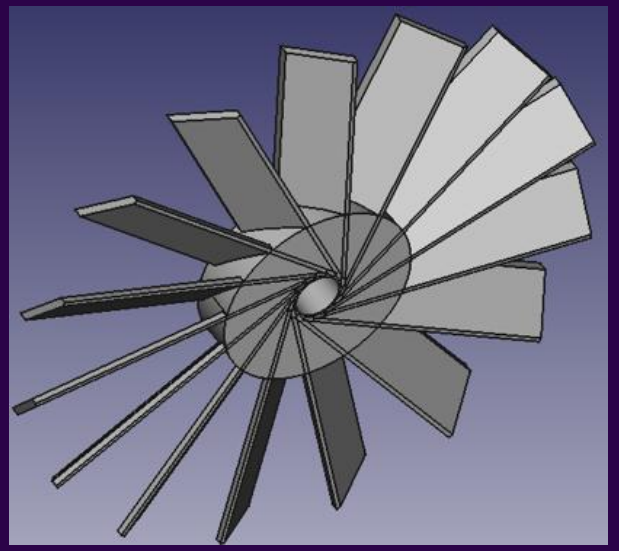
Comprehensive understanding of the technical requirements of motorsport



SCHOOL AND SELF PROJECTS

CREATE A MINI WIND TUNNEL







- Design and construction of a mini wind tunnel for measuring lift on a simple wing profile
- Sizing of the acquisition system with piezoelectric force sensor and speed measurement via anemometer
- CAD design and additive manufacturing of NACA 4412 profiles
- Objective: to correlate flow velocity and experimentally measured lift force.



2

REDESIGN A RANGE OF ROAD TRAILERS

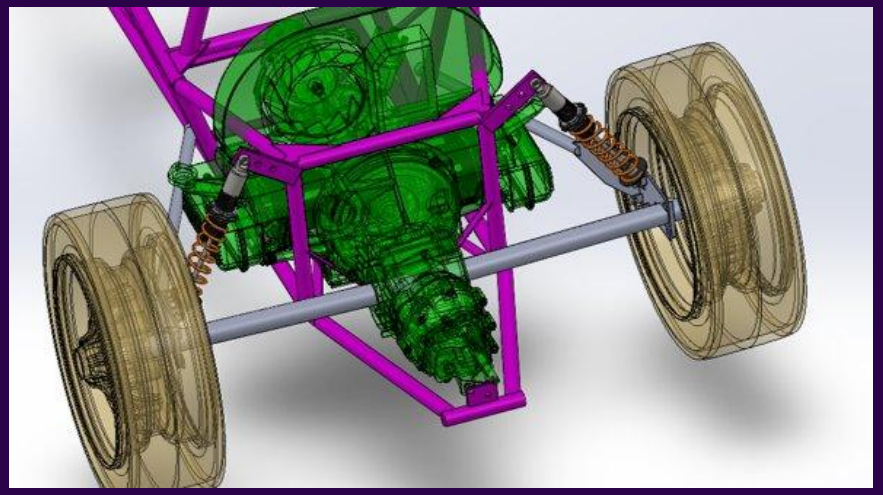


- Industrial project in collaboration with LIDER aimed at reducing weight and costs while improving product durability
- Materials study:
  - Polyurethane and honeycomb panels
  - Mechanical validation up to 27 MPa (bending and impact testing)
- Structural optimization:
  - Galvanised S235 steel
  - S700 cataphoretic steel
- Development:
  - Load gauge
  - Integrated side box
  - Panel removal system

For reasons of confidentiality, I have not been authorized to provide any other values, photos or 3D models.

SIZING OF A REAR SUSPENSION FOR A FORMULA VEE

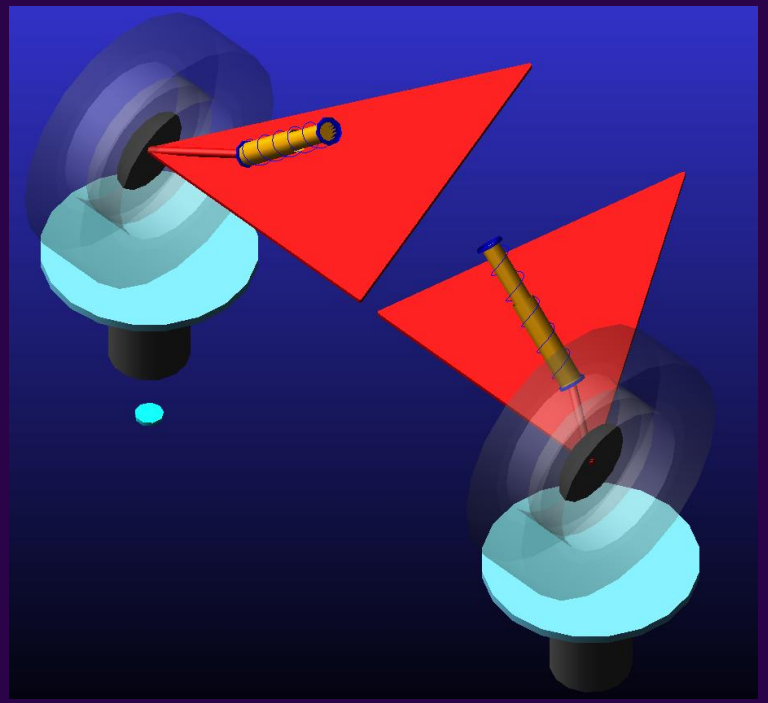


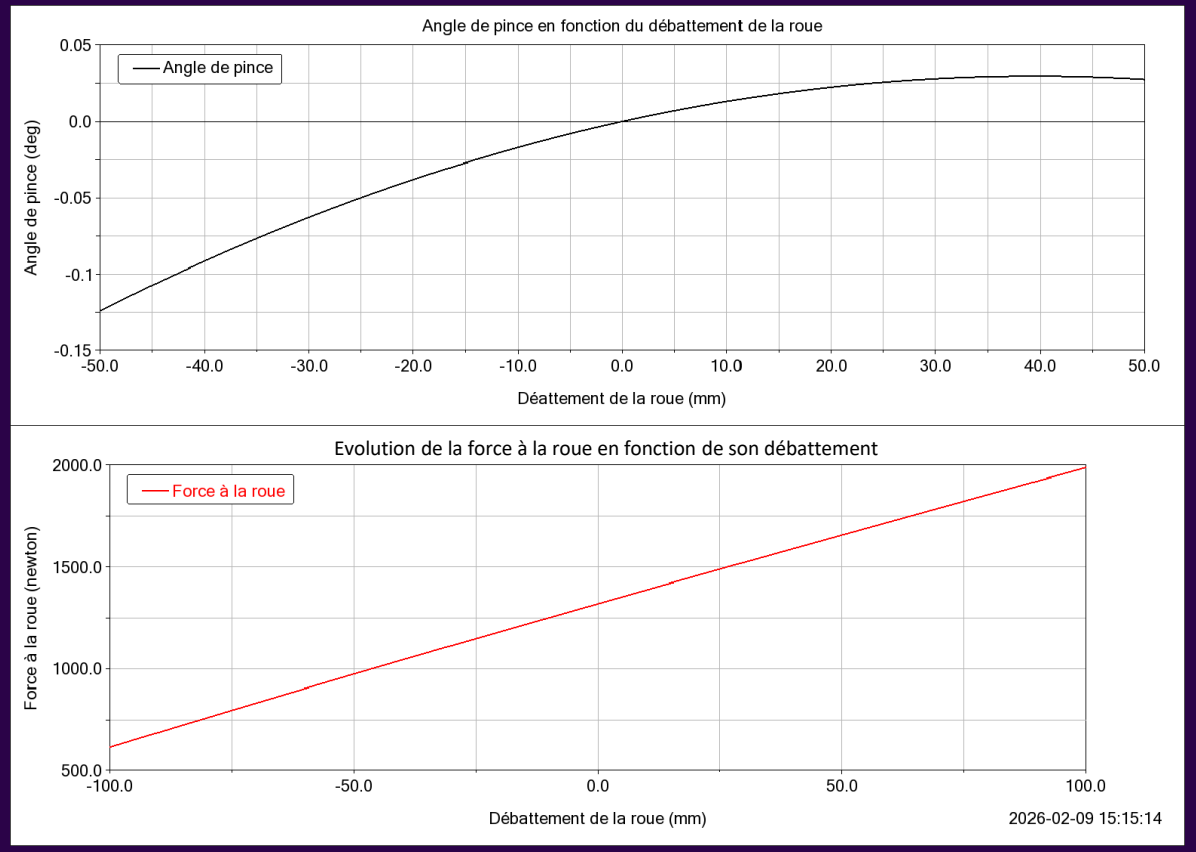


- Sizing Formula Vee rear suspension as part of a vehicle renovation project
- Architecture choice based on cost / constraints
- Dynamic calculations:
  - Motion ratio: 0.52
  - Spring stiffness: 35 N/mm
  - Wheel stiffness: 8 N/mm
- Validation:
  - Dynamic simulation using ADAMS Car
  - Frequency verification using Matlab Simulink
- Kinematics analysis:
  - Toe-in
  - Camber
  - Roll behaviour

SIZING OF A REAR SUSPENSION FOR A FORMULA VEE

- Lower triangle fixed to the hub => single-piece modelling
- Combined ball joint on both sides



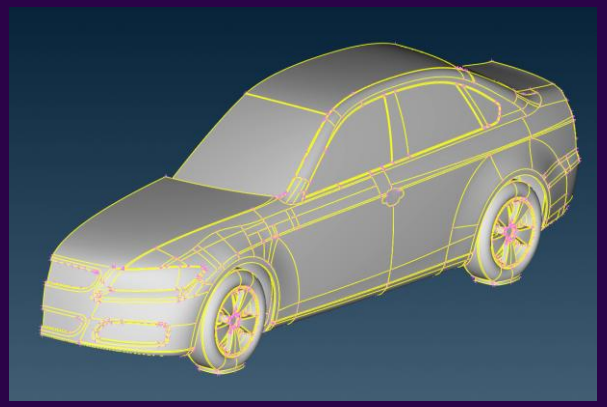


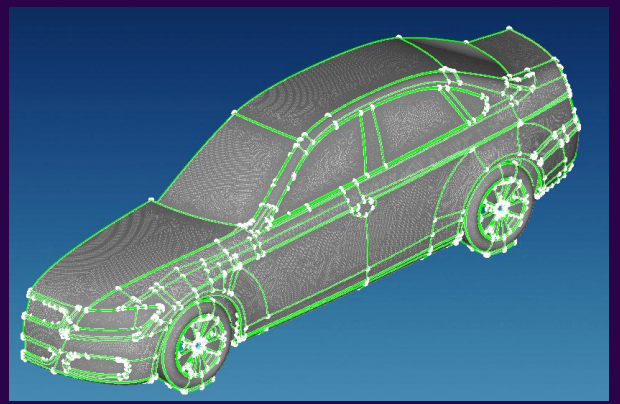
CFD SIMULATION OF A DRIVAER CAR MODEL

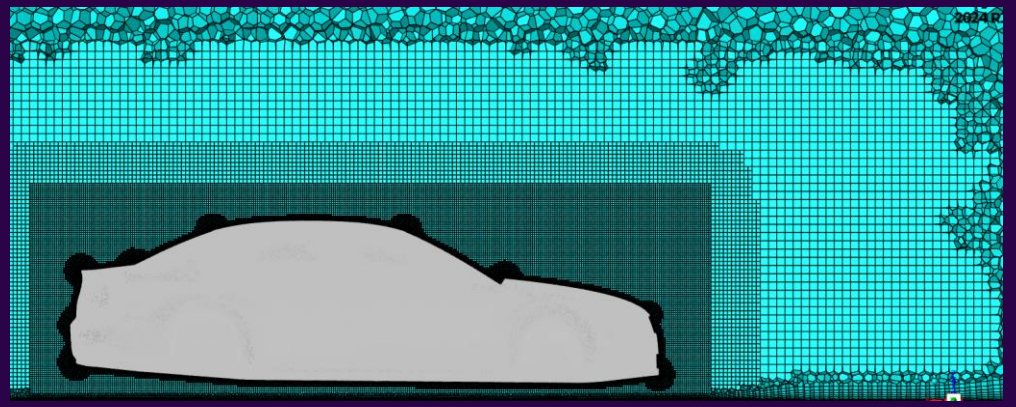
- Project to perform CFD simulations on a surface model of a car

- CAD cleaning:
  - Ansa
  - File cleaning
  - Integration of the fluid domain and BOIs

- Maillage:
  - Refined surface mesh
  - Script and Strategic volume mesh



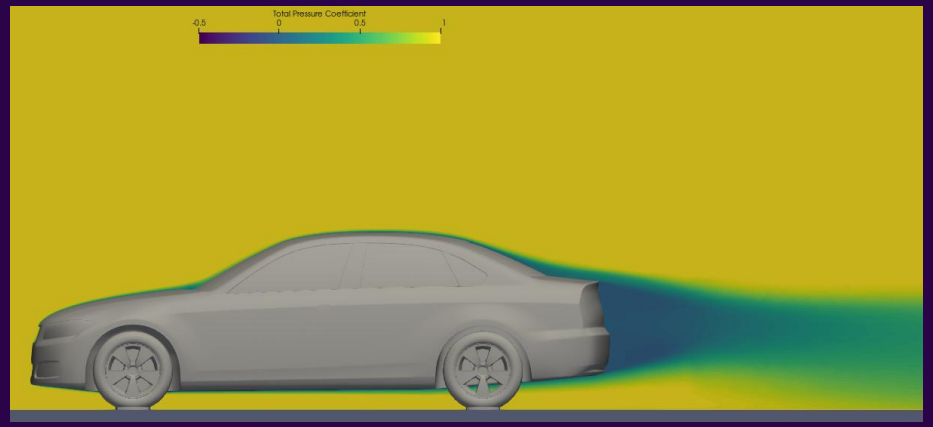


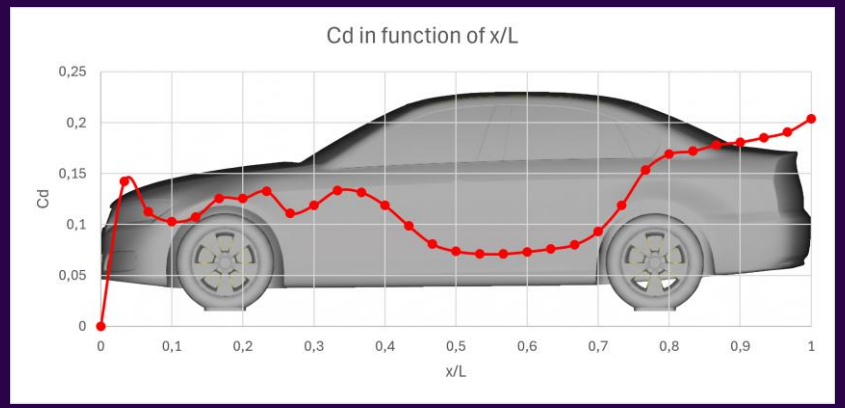


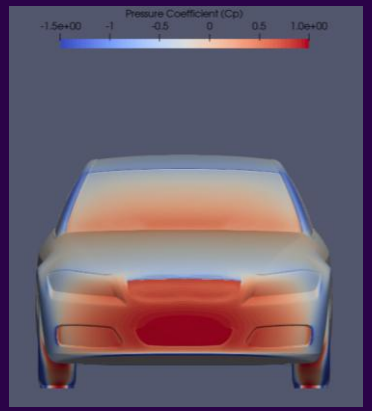
CFD SIMULATION OF A DRIVAER CAR MODEL

- CFD simulation on a HPC
  - Fluent
  - HPC (Linux)
  - k epsilon realizable with enhanced-wall-treatment
  - Solving script and Simulation with rotating wheels

- Analysis of results:
  - Use ParaView for automatic processing
  - Average Cd ≈ 0,21









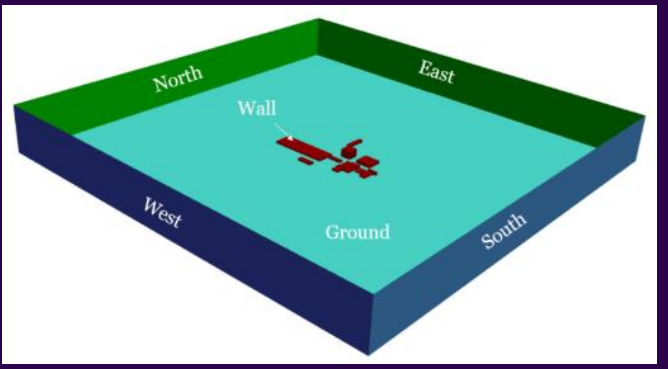
CFD SIMULATION OF A FIRE IN AN OPEN AREA

- Projet to perform CFD simulations of a fire on the Valla campus of Linköping University

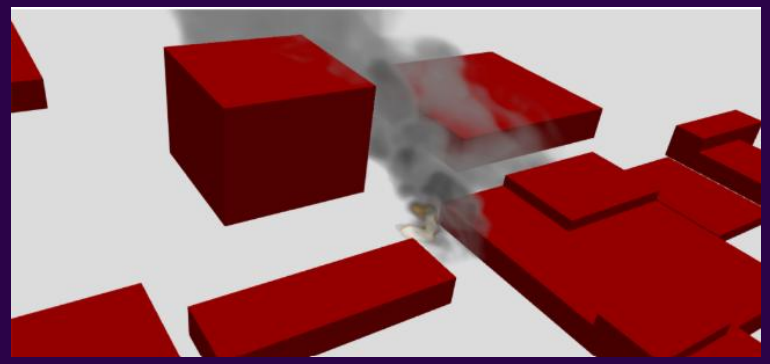
- CFD Simulation
  - Fluent
  - HPC (Linux)
  - k epsilon realizable with enhanced-wall-treatment
  - Simulation with chemical reaction and passive scalar
  - Solving cript and use of the energy equation

- Analysis of results
  - Use of ParaView for automatic processing

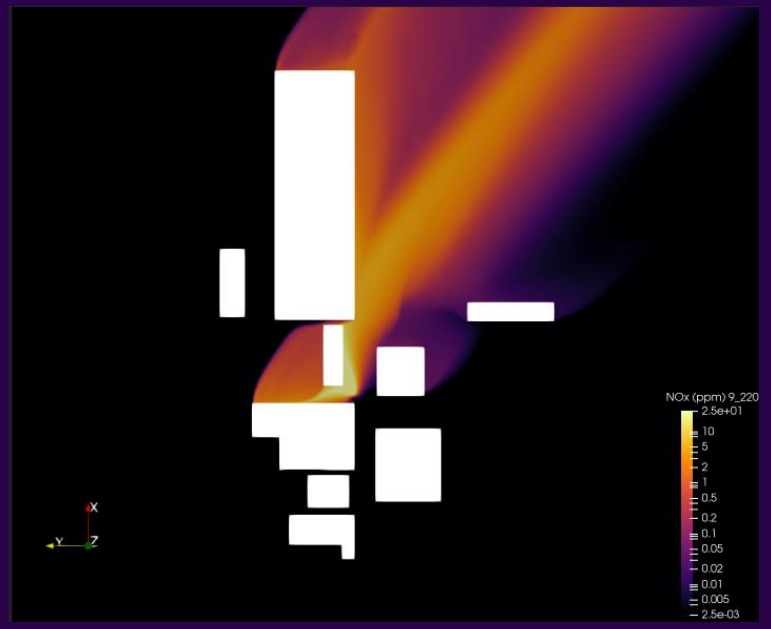




CFD SIMULATION OF A FIRE IN AN OPEN AREA



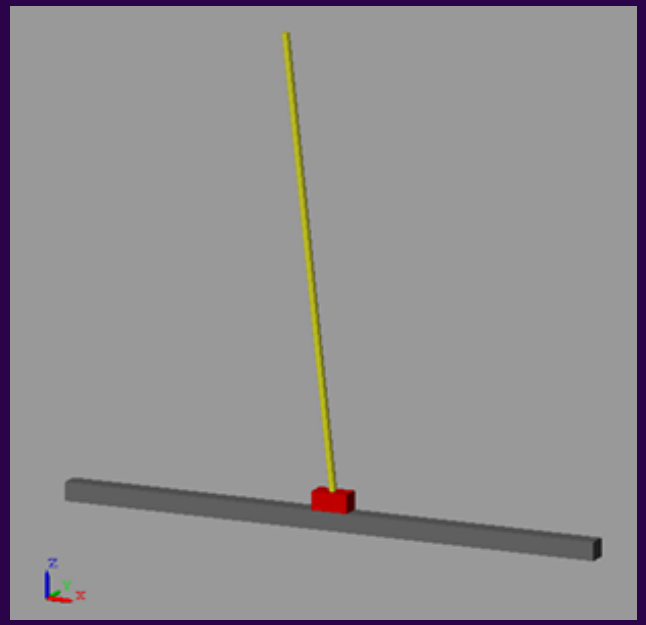
Simulation with chemical reaction

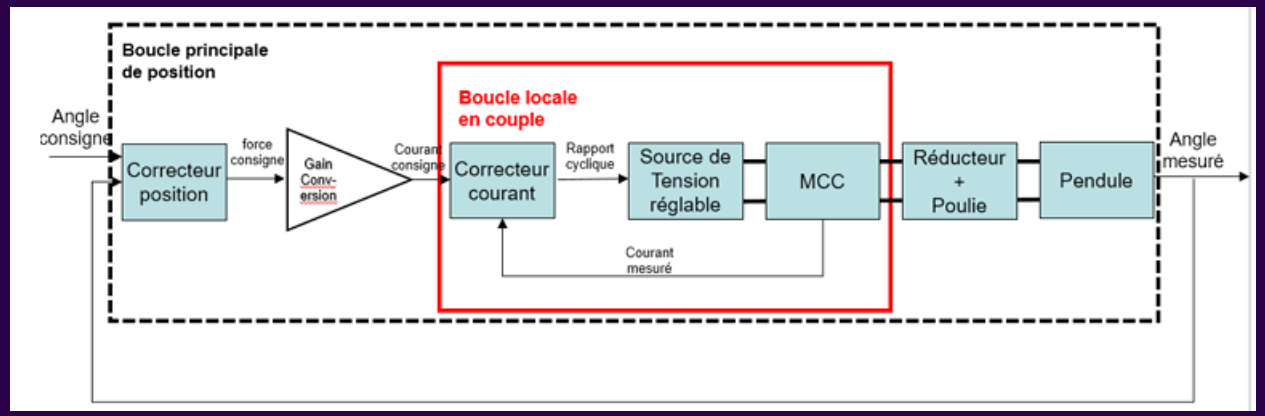


Simulation with passive scalar



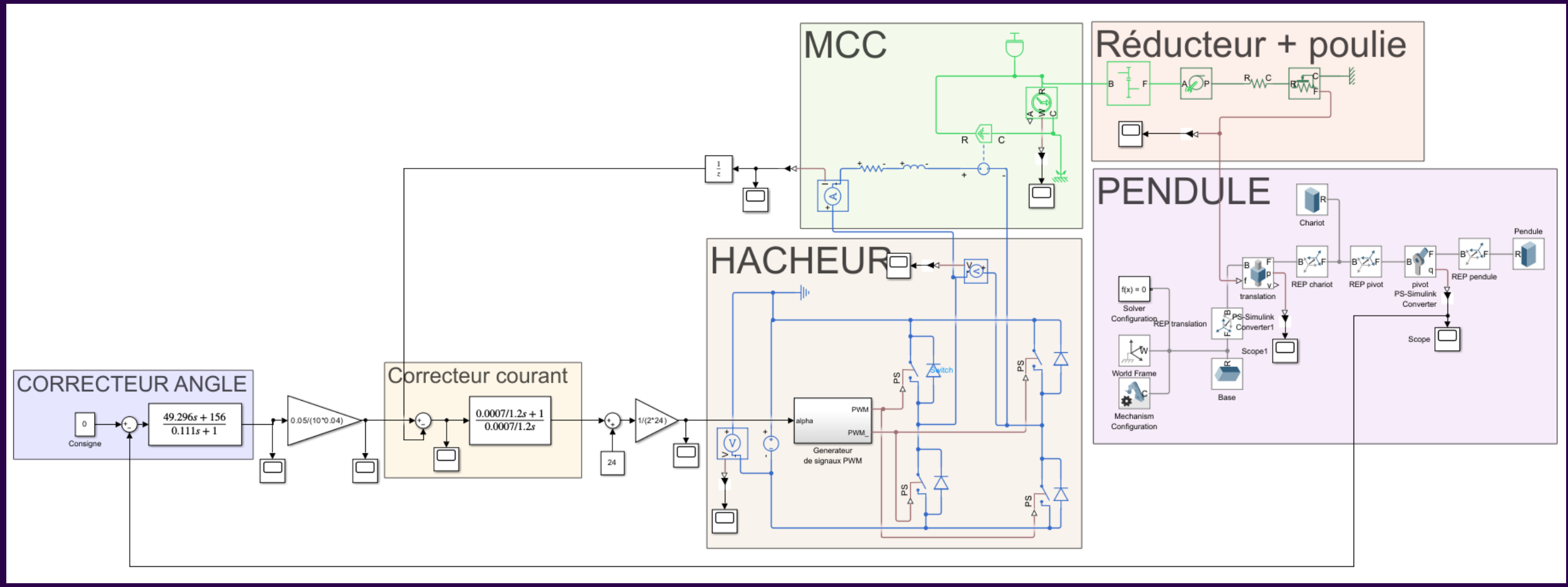
CREATING THE CONTROL FOR AN INVERTED PENDULUM



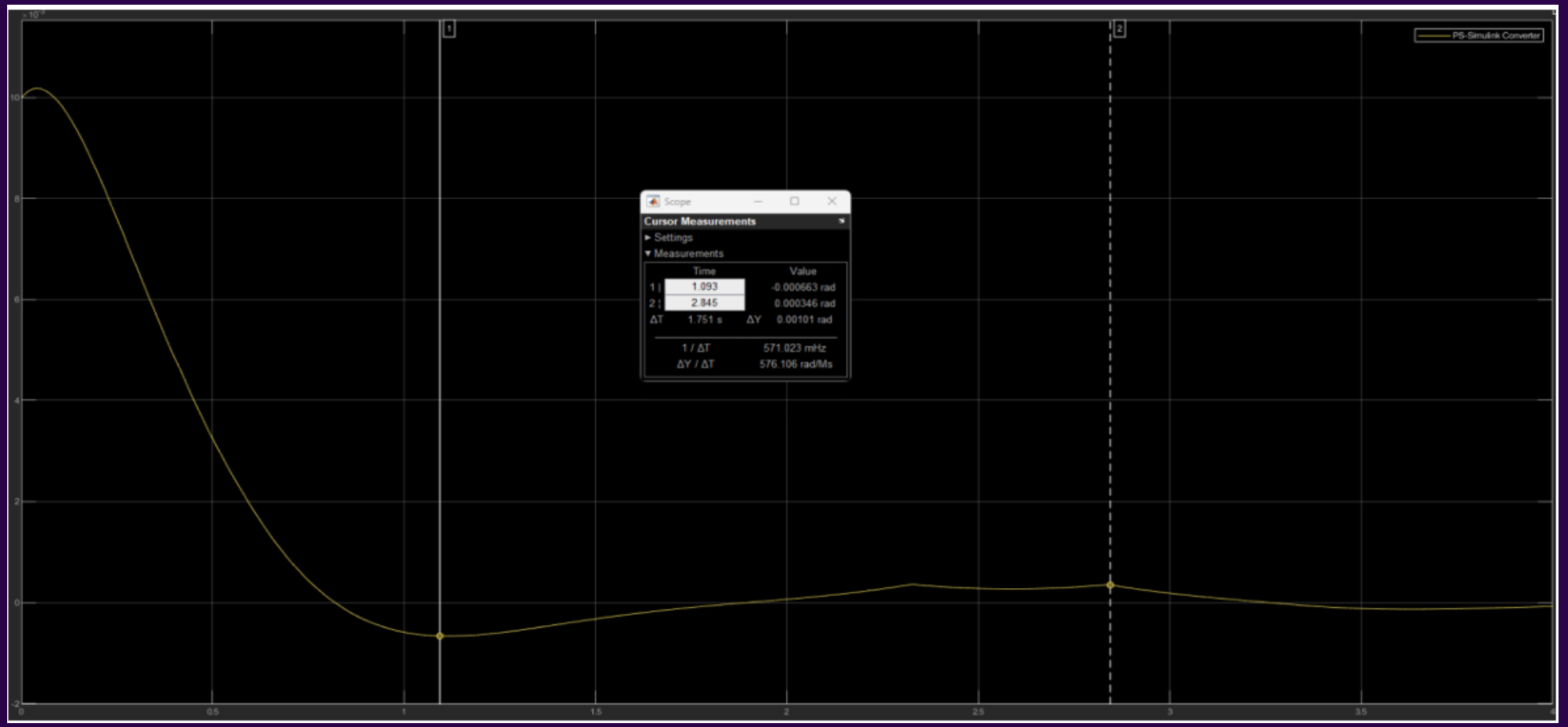


- Design of a control system for an inverted pendulum with correction
- Modeling of a chopper and an electrical machine
- SIMSCAPE modeling of the pendulum
- Insertion of current and angle controllers
- Analysis:
  - -Good operation of the chopper
  - -System voltage
  - -Motor rotation speed
  - -Angular position and carriage position as a function of time

CREATING THE CONTROL FOR AN INVERTED PENDULUM



CREATING THE CONTROL FOR AN INVERTED PENDULUM



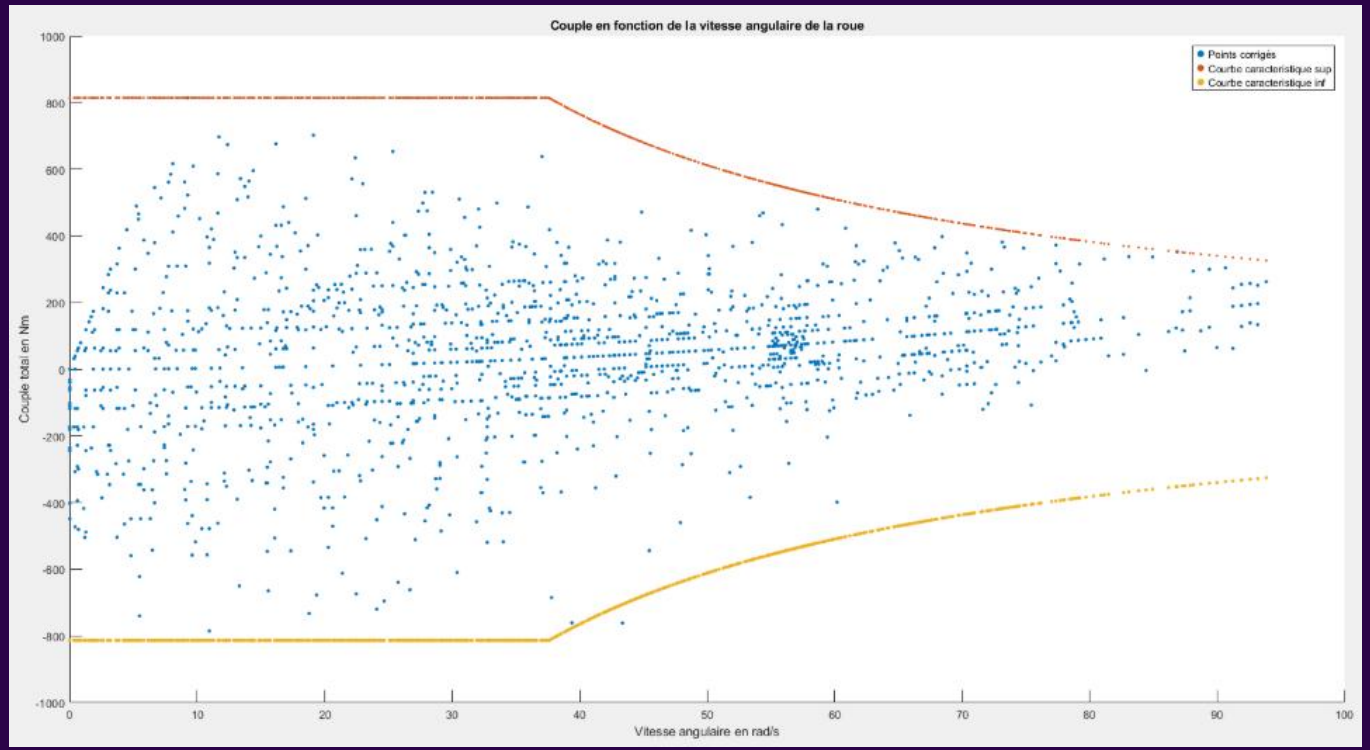
SIZING A BATTERY PACK 1/3

- Sizing a motor for an electric vehicle
- Input data: mixed ARTEMIS cycle and vehicle data
- Selects the appropriate motor type based on:
  - The minimum power required to cover the least achievable point (maximum torque)
  - in generator and motor modes

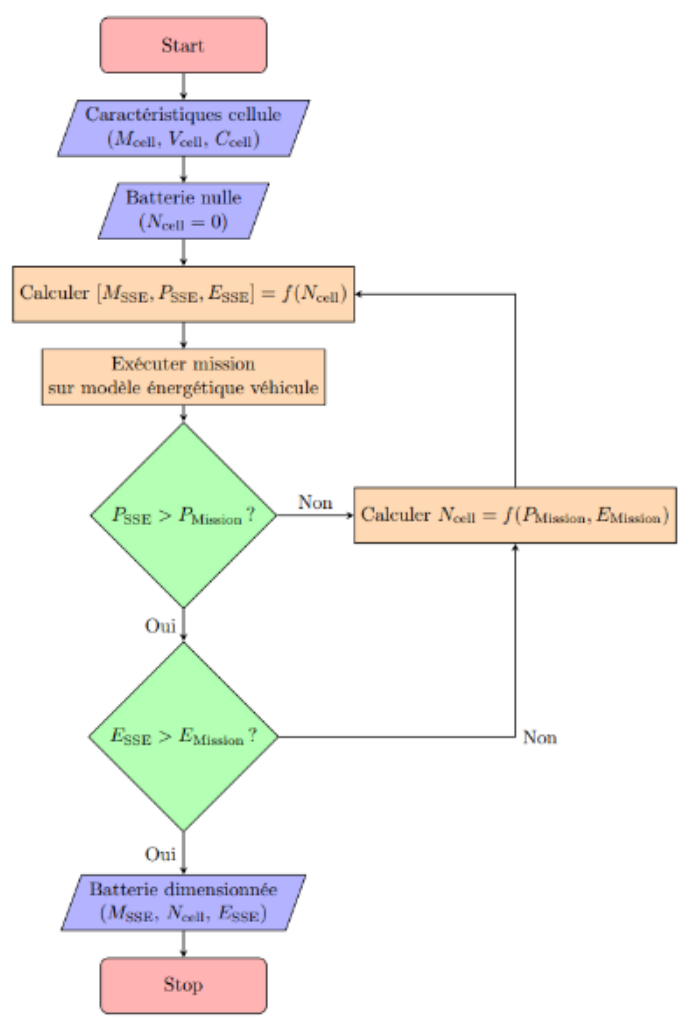
$$\begin{cases} F_{\text{aero}} &= \frac{1}{2} \cdot \rho \cdot s \cdot C_x \cdot V_{\text{veh}}^2 \\ F_{\text{roue}} &= M_{\text{veh}} \cdot g \cdot C_{rr} \cdot V_{\text{veh}}^2 \\ F_{\text{gx}} &= M_{\text{veh}} \cdot g \cdot \sin(\alpha) \end{cases}$$

onc exprimées en fonction des masses du VE. Ainsi, la pu
ut être exprimée de nouveau par :

$$P_V = \left(M_{\text{veh}} \cdot \frac{\mathrm{d}V_{\text{veh}}}{\mathrm{d}t} + F_{\text{aero}} + F_{\text{roue}} + F_{\text{gx}} \right) \cdot V_{\text{veh}}$$



SIZING A BATTERY PACK 2/3



- Sizing a traction battery for an electric vehicle
- MATLAB Simulink modeling of the block diagram to determine the number of cells, number of battery packs, energy capacity, and total cost
- Automatic selection of the best cell technology from several available options
- Limiting the depth of discharge to 80%
- Consideration of packaging representing 30% of each cell's mass
- Calculation of the number of cells per branch and the number of branches for branches up to 120V.

SIZING A BATTERY PACK 2/3

```
for k = 1:Ntypes

    Mcell   = batt(k,1);
    Vcell    = batt(k,2);
    EnergSpe  = batt(k,3)*3600;
    Id      =   batt(k,4);
    PrixCell  = batt(k,5);

    Ncell = 0;
    i = 1;

    while i >= 1

        MSSE=Ncell*Mcell;
        ESSE=EnergSpe*MSSE;
        PSSE=Vcell*Id*Ncell;
        sim('TP1Simu.slx');
        p=Puissance;
        e=Energie;

        if PSSE<max(p)
            Ncell=ceil(max(max(p)/(Vcell*Id),max(e)/(EnergSpe*Mcell)));
        elseif ESSE<max(e)
            Ncell=ceil(max(max(p)/(Vcell*Id), max(e)/(EnergSpe*Mcell)));

        else
            i = 0;
        end
    end


    results(k).Ncell = Ncell;
    results(k).MSSE  = Ncell * Mcell;
    results(k).ESSE  = EnergSpe * Ncell * Mcell;
    results(k).Cout  = Ncell * PrixCell;
end

Couts = [results.Cout];
[minCost, idBest] = min(Couts);

disp(['Type batterie        : ', num2str(idBest)]);
disp(['Nombre cellules      : ', num2str(results(idBest).Ncell)]);
disp(['Masse totale (kg)    : ', num2str(results(idBest).MSSE)]);
disp(['Energie totale (J)   : ', num2str(results(idBest).ESSE)]);
disp(['Coût total (€)       : ', num2str(results(idBest).Cout)]);
```

```
data=load('ArtemisMIXTE.mat');



Mveh=860;


s=2.75;
rho=1.2;
Cx=0.3;
Crr=1.6e-5;
R=0.33;
alpha=0;

%  [Mcell  Vcell  EnergSpe(Wh/kg)  Courant  PrixCell]

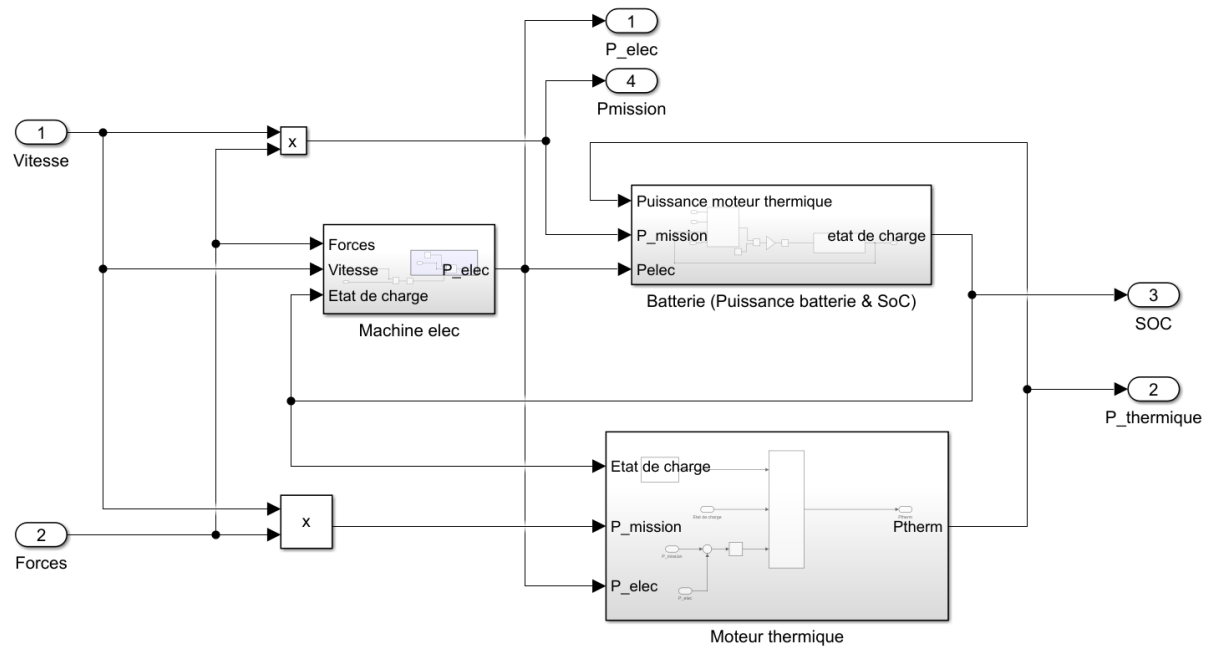
batt = [
    1.03   3.7   143.68   200   82.22;   % Batterie 1
    2.5    3.7   148.00   500   209.9;   % Batterie 2
    0.940  3.65   128.00   500   67.8;   % Batterie 3
    0.885   3.7  167.25   40    76.33;   % Batterie 4
    2.250   3.7   164.44   100   190.81;   % Batterie 5
    0.9   3.3   155.55   50    72.24;    % Batterie 6
];


VvehTab = data.ArtemisMIXTE(:,2);

Ntypes = size(batt,1);

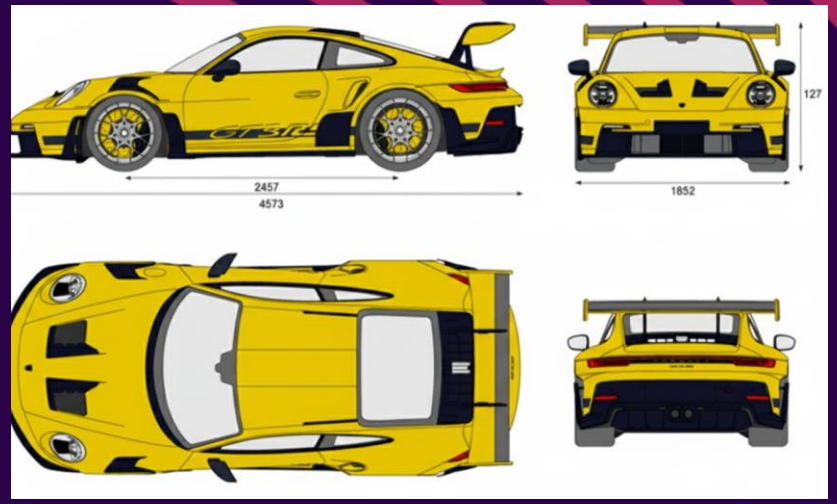

results = struct('Ncell',[],'Mtot',[],'Etot',[],'Cout',[]);
```

SIZING A BATTERY PACK 3/3



Moteur	Batterie	Pnom (kW)	Masse (kg)	Capacité (Ah)	Tension (V)	Degré hybridation	Temps thermique (s)	Classification
A1	U12	5	15	20	12	0,6762	3911	Full parallèle
	U48		50	25	48	0,6737	3941	
	U400		200	20	400	0,6498	4230	
B2	U12	29	15	20	12	0,978	266	Range Extender
	U48		50	25	48	0,9751	301	
	U400		200	20	400	0,9657	414	
C3	U12	150	15	20	12	0,9779	267	
	U48		50	25	48	0,9779	267	
	U400		200	20	400	0,9768	280	
D4	U12	50	15	20	12	0,9779	267	
	U48		50	25	48	0,9779	267	
	U400		200	20	400	0,9768	280	

- Sizing a hybrid powertrain
- Implementation of terminals, Pmax and -Pmax of the motor due to power limitations imposed by the battery
- State of Charge (SoC) settings
- Insertion of a switch to return 0 motor power when the battery is charging. When discharging, the motor power is displayed.
- Determination of the internal combustion engine power. Electric power when the vehicle is under load.
- Calculation of the internal combustion engine's operating time and the vehicle's degree of hybridization.



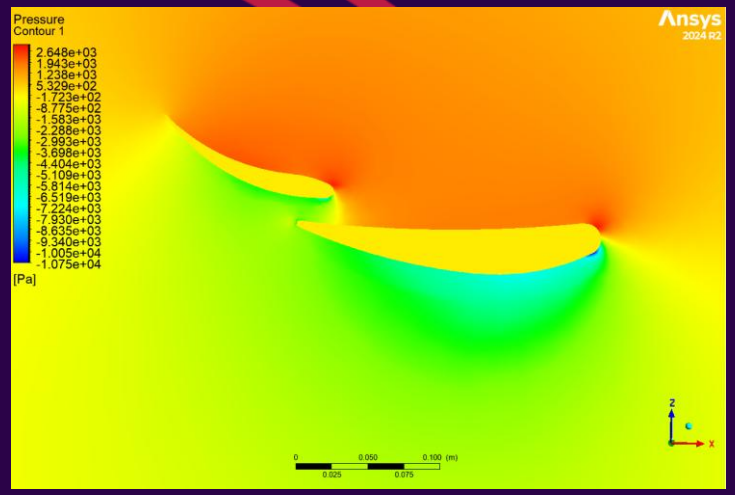


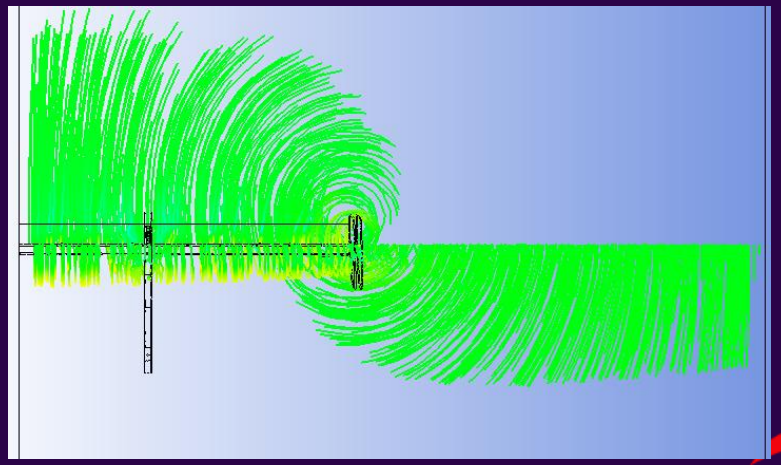


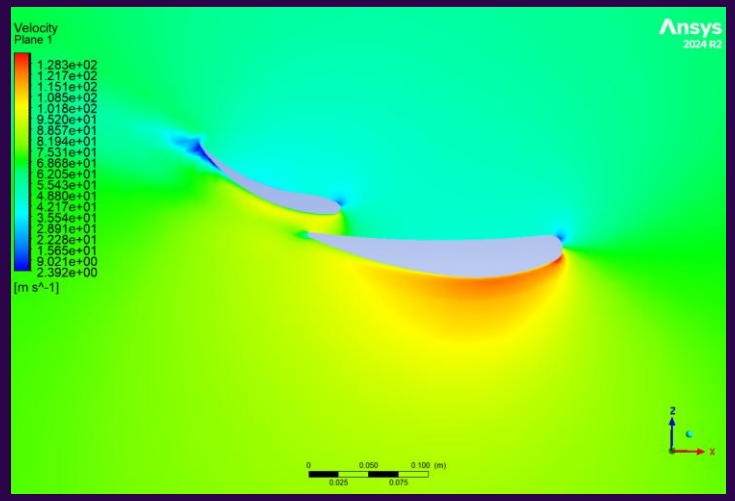
CFD TEST OF A PORSCHE 911 992 GT3 RS WING

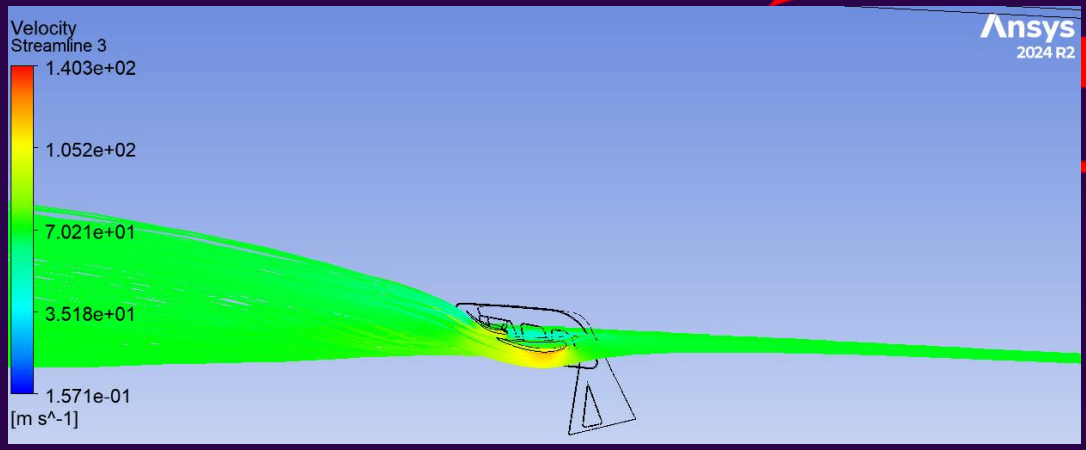
- CFD study of a GT3-type spoiler carried out as a personal project
- Steps:
  - CAD modelling in 3DEXPERIENCE based on manufacturer's plans
  - CFD simulation in ANSYS Fluent at 70 m/s
- Analysis:
  - Lift
  - Drag
  - Pressure distribution
- Validation: qualitative comparison with manufacturer's data

CFD TEST OF A PORSCHE 911 992 GT3 RS WING



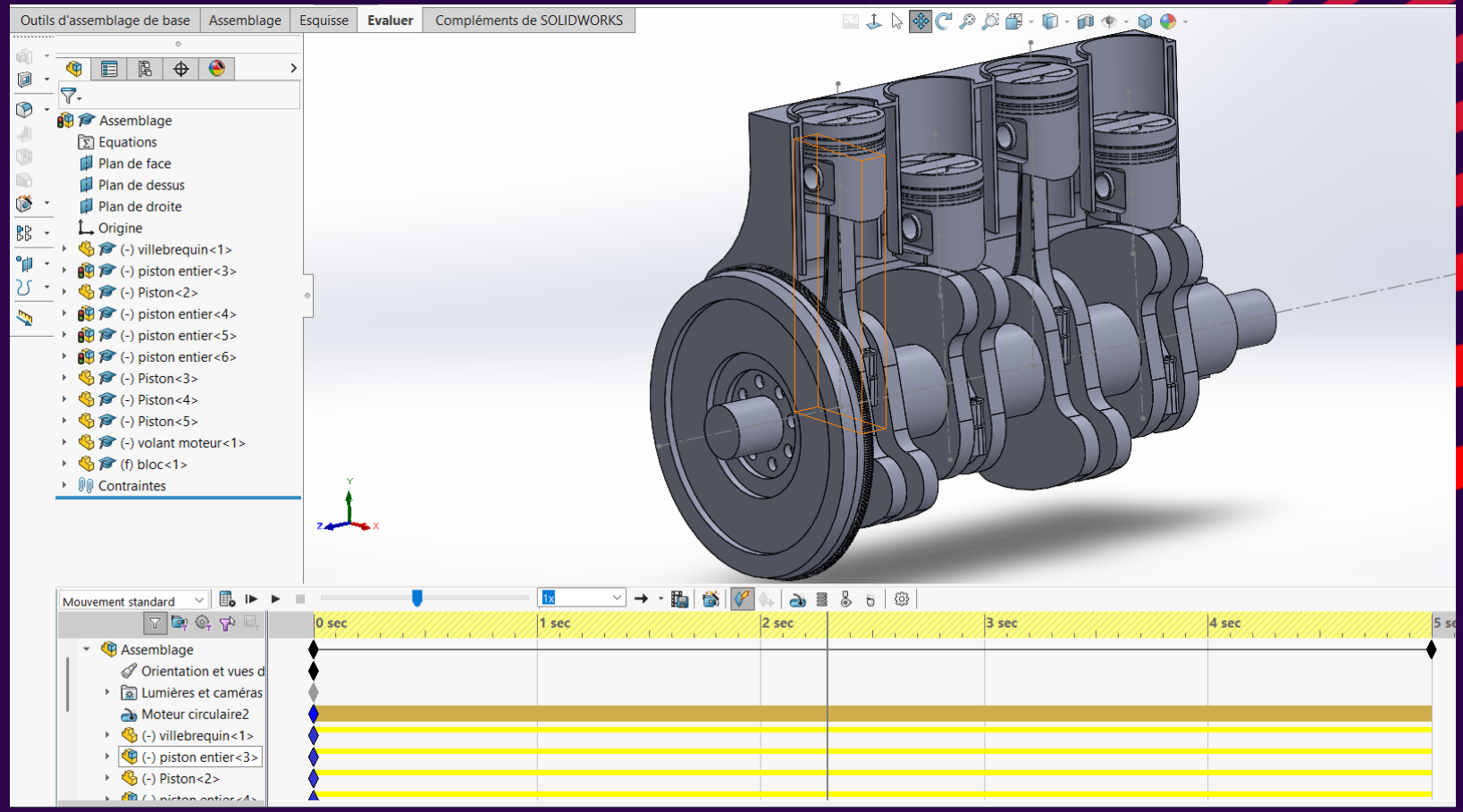




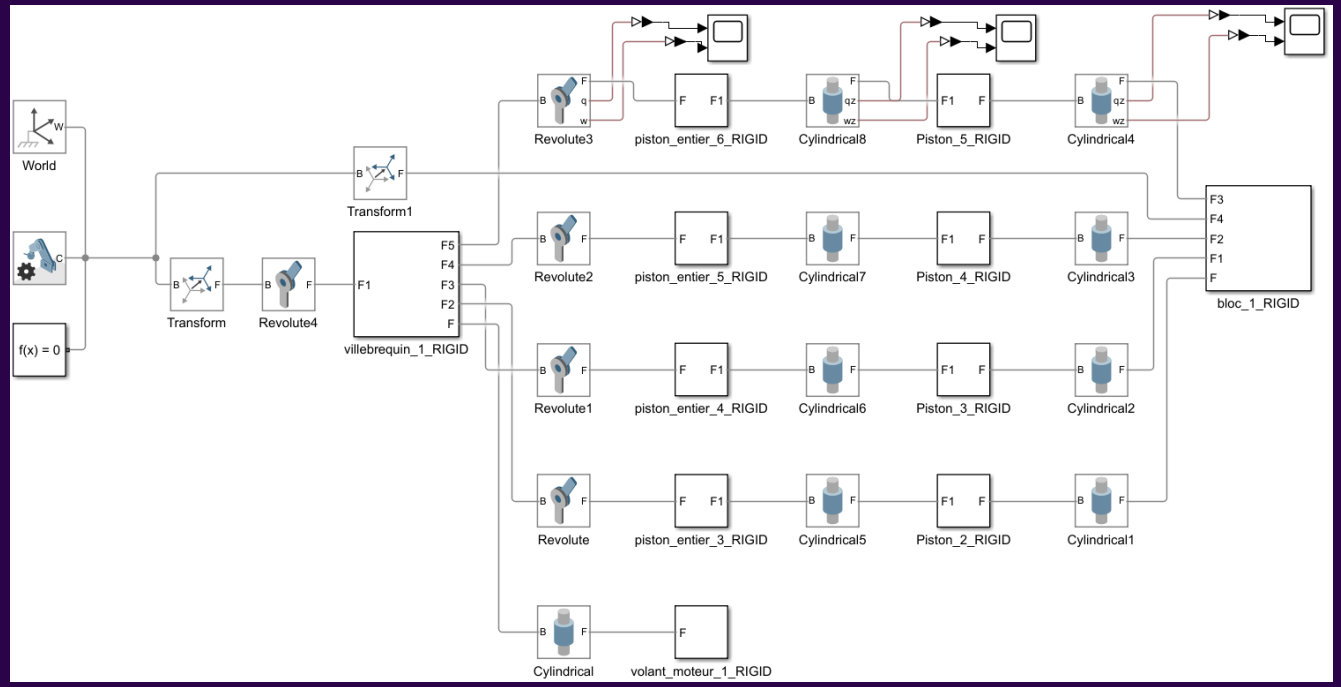


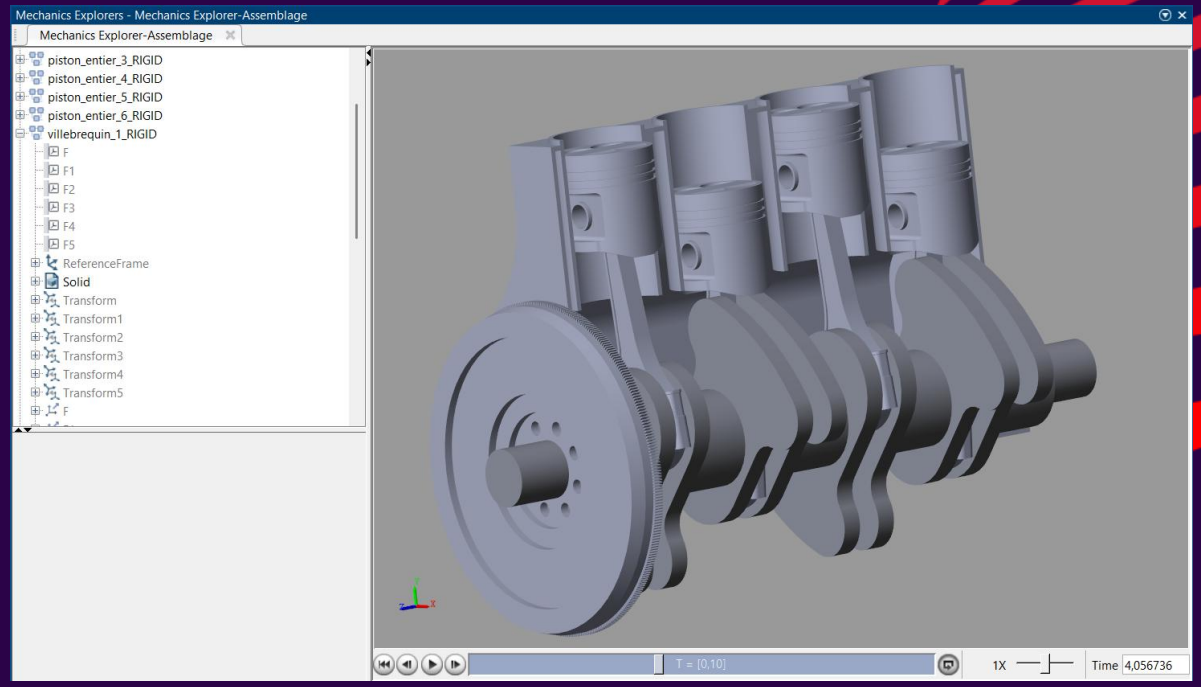
ANALYSING THE MOVEMENT OF AN ENGINE

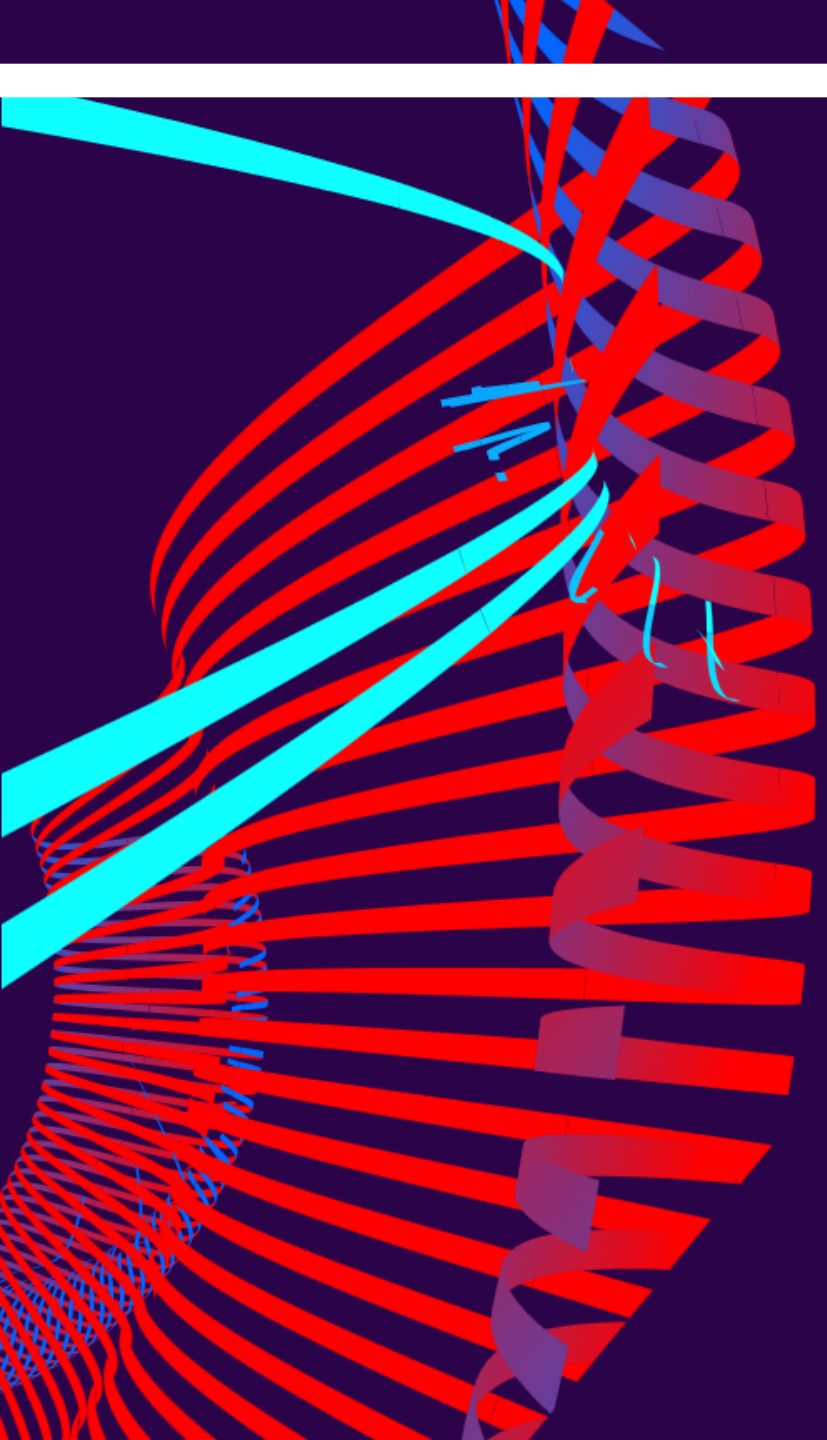
- Engine Rotation Modeling and Analysis
- SOLIDWORKS 3D Modeling: 4-Cylinder Inline Engine - Automatic Rotation
- SOLIDWORKS to SIMULINK
  - -Upload Package
  - -Automatic Modeling of Blocks and connections (SIMSCAPE)
  - -Crankshaft Rotation at 100°/s
  - -Analysis of Piston and Connecting Rod movement



ANALYSING THE MOVEMENT OF AN ENGINE







INTERNSHIPS

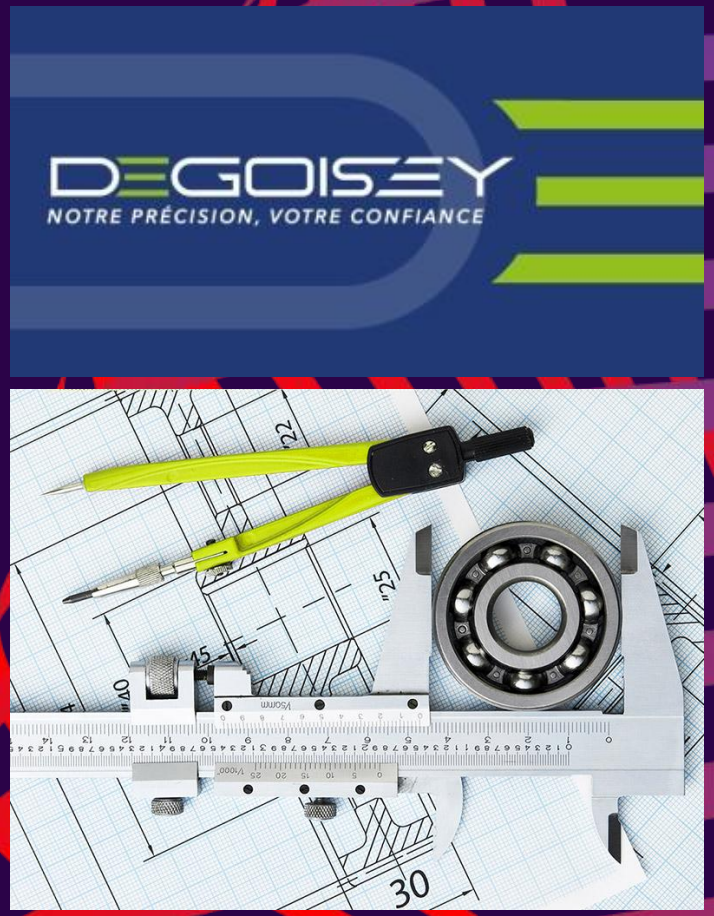
METROLOGY INTERNSHIP

- Metrological inspection of machined parts for customer compliance validation

- Use:

  Manual metrology instruments
  Three-dimensional measuring machines
  Imaging measurement
  Contourograph

- Development of strict tolerances and industrial plan reading



AUTOMOTIVE MECHANICS INTERNSHIP



- Complete immersion in vehicle architecture and on-board mechanical and electronic systems

- Work on:
  - Powertrains
  - Transmissions
  - Running gear
  - On-board electrical systems

- Learning:
  - Technical diagnostics
  - Interaction between mechanical and electronic systems
  - Vehicle maintainability constraints

11

DESIGN INTERNSHIP FOR AUTOMOTIVE TESTING

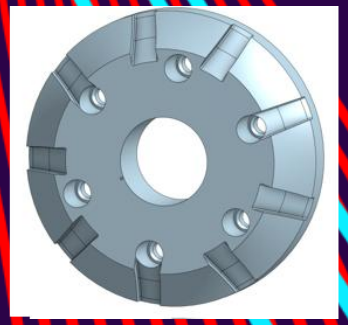
- Design of a test bench and a Brake-by-Wire system & Sizing of an air extractor
- Design and 3D modeling of a braking fixing system and a rim trainer based on a 3D scan carried out
- Carry out automotive braking tests
- Study, sizing and integration of a Brake-by-Wire braking system
- Pressure, deformation (caliper), force and displacement measuring
- Participation in automobile tests and experimental validation
- Thermal analysis and verification of the resistance of materials in the sizing of the air extractor
- Adaptation and reuse of existing automotive infrastructures and technologies (AVL puma bench, test infrastructures, Stähle Robot autopilot)

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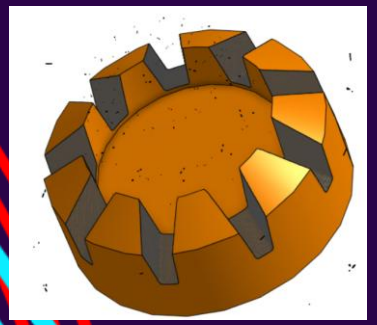
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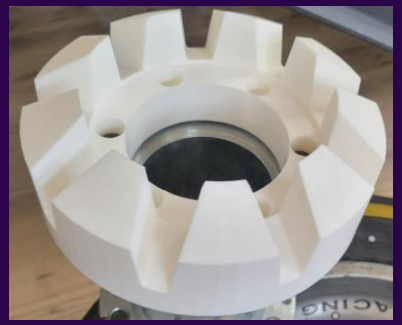
Creating a rim drive to rotate the wheel

From a 3D file



Using a real rim and a 3D scan



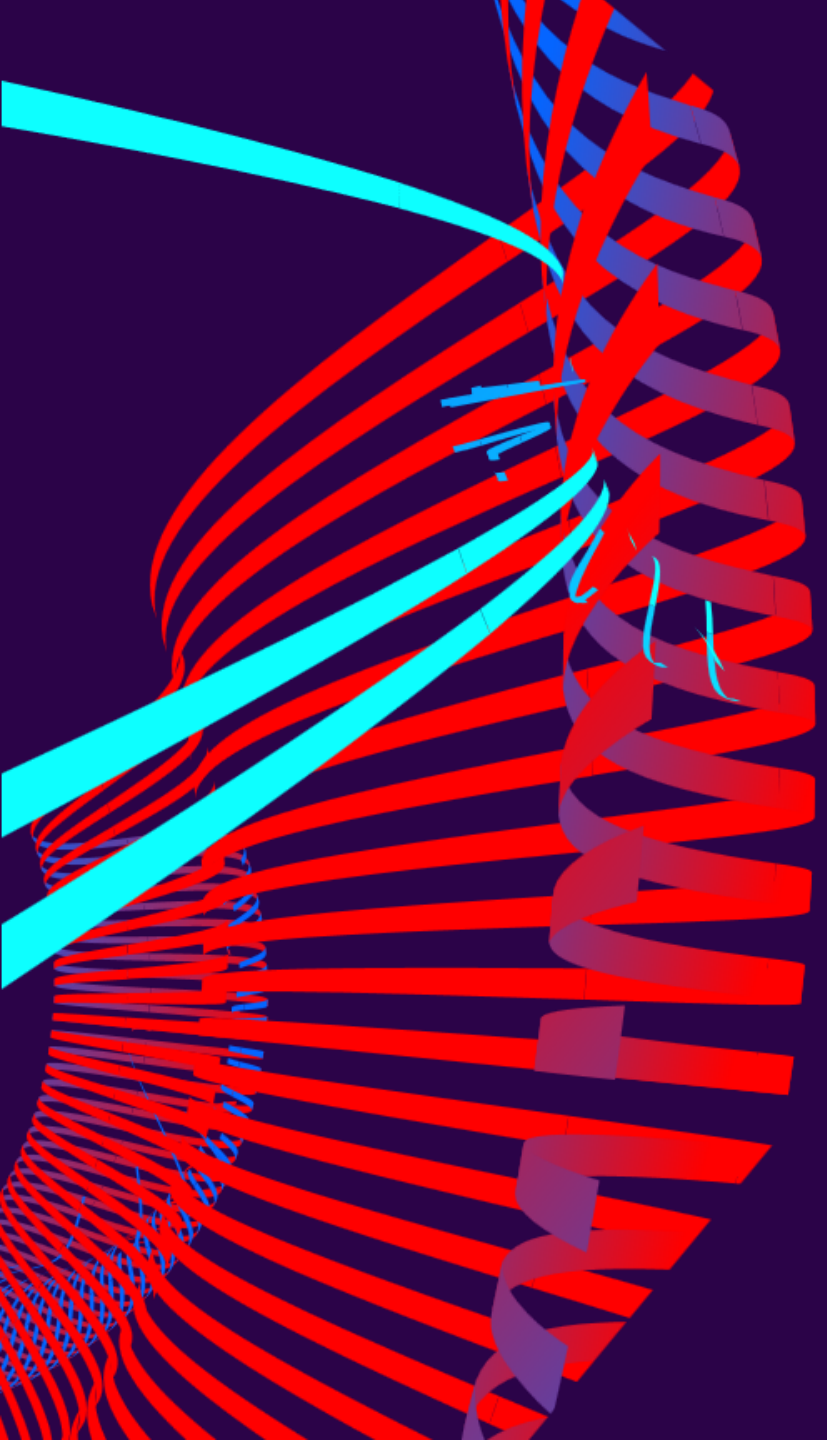


Construction of a support to connect a brake master cylinder and the pilot robot

Creation of a support with a roller to pre-stress the tire and reproduce braking conditions

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For reasons of confidentiality, I have not been authorized to provide any other values, photos or 3D models.



AND NOW ?

- looking for a 6-month **internship** in automotive engineering in R&D from **February 1st to Jully 2027 (6 month)** in:

  - ✓ Automotive Industry or motorsport
  - ✓ Design office – R&D
  - ✓ CAD - CFD
  - ✓ Automotive design / mechanics
  - ✓ Anywhere in the world

MY INTERESTS

Automobile and motorsport

Motorsport enthusiast since I was young, especially by Formula One and Rally

Track and field

Half-Professionnal athlete in 400m and 400m hurdles

THANKS

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